

TARGET : PRE-MEDICAL 2020 ENTHUSIAST & LEADER COURSE

PHASE : ALL

Test Type : MAJOR

Test Pattern : NEET(UG)

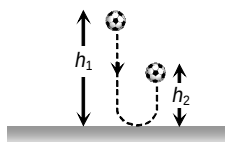
TEST DATE : 19 - 03 - 2020

TEST SYLLABUS : 03 & 04

HINT - SHEET

1. According to energy conservation, change in potential energy of the ball, appears in the form of heat which raises the temperature of the ball.

i.e. $mg(h_1 - h_2) = m.c.\Delta\theta$

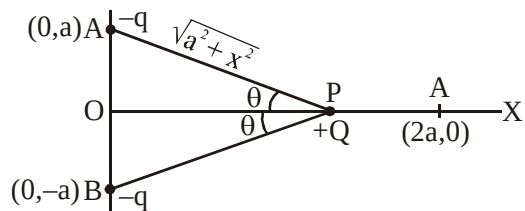


$$\Rightarrow \Delta\theta = \frac{g(h_1 - h_2)}{c}$$

$$= \frac{10(10 - 5.4)}{460} = 0.1^\circ\text{C}$$

3. Time = $\frac{\text{Distance}}{\text{Velocity}} = \frac{1000}{330} = 3.03 \text{ sec.}$

Sound will be heard after 3.03 sec. So his watch is set 3sec, slower.

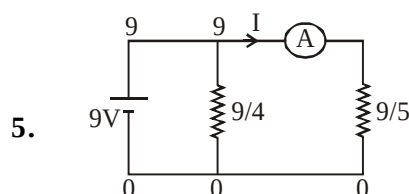
4. 

Component of force on charge +Q at P, along x-axis,

$$F_x = \frac{2Qq}{4\pi\epsilon_0(a^2 + x^2)} \cos\theta$$

$$= \frac{2Qq}{4\pi\epsilon_0(a^2 + x^2)} \times \frac{x}{\sqrt{a^2 + x^2}}$$

$$= \frac{2Qqx}{4\pi\epsilon_0(a^2 + x^2)^{3/2}}$$

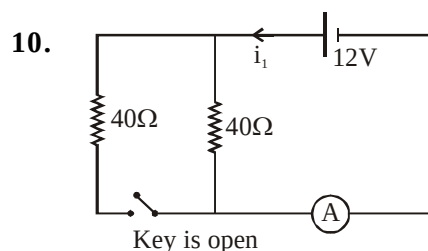


$$I = \frac{9-0}{9/5} = 5\text{A}$$

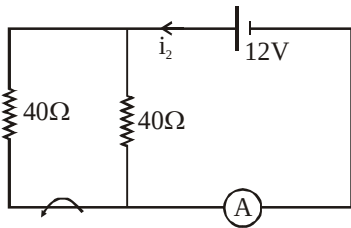
6. Change in internal energy (ΔU) depends upon initial and final state of the function while ΔQ and W are path dependent also.

8. $v = \sqrt{\frac{K}{\rho}} \therefore K = v^2 \rho = 2.86 \times 10^{10} \text{ N/m}^3$

9. We know that magnitude as well as polarities of charges at points A and C are same. Therefore, forces at O, due to these charges cancel each other. Moreover, polarities of the charges at B and D are opposite; therefore force on charge kept at the centre is along the diagonal BD.



$$i_1 = \frac{12}{40} = \frac{3}{10} \text{ A}$$



$$i_2 = \frac{12}{20} \Rightarrow i_2 = \frac{6}{10} \Rightarrow \frac{i_1}{i_2} = \frac{3 \times 10}{10 \times 6} = \frac{1}{2} = 0.5$$

11. $W = \mu RT \log_e \frac{V_2}{V_1}$

$$= \left(\frac{m}{M} \right) RT \log_e \frac{V_2}{V_1} = 2.3 \times \frac{m}{M} RT \log_{10} \frac{V_2}{V_1}$$

$$= 2.3 \times \frac{96}{32} R (273 + 27) \log_{10} \frac{140}{70} = 2.3 \times 900 R \log_{10} 2$$

13. The tuning fork whose frequency is being tested produces 2 beats with oscillator at 514 Hz, therefore, frequency of tuning fork may either be 512 or 516. With oscillator frequency 510 it gives 6 beats/sec, therefore frequency of tuning fork may be either 516 or 504.

Therefore, the actual frequency is 516 Hz which gives 2 beats/sec with 514 Hz and 6 beats/sec with 510 Hz.

14. Electric lines are perpendicular to equipotential surface.

15. Internal resistance

$$r = \left(\frac{\ell_1}{\ell_2} - 1 \right) R = \left(\frac{50}{40} - 1 \right) \times 2 = 0.50 \Omega$$

16. $PV^\gamma = \text{constant}$

$$\Rightarrow \frac{P_2}{P_1} = \left(\frac{V_1}{V_2} \right)^\gamma \Rightarrow P_2 = (8)^{5/3} P_1 = 32 P_1$$

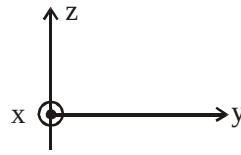
17. Thermal conductivity of composite plate

$$K_{eq} = \frac{2K_1K_2}{K_1 + K_2} = \frac{2 \times 2 \times 3}{2 + 3} = \frac{12}{5} = 2.4$$

18. $n \propto \sqrt{T}$

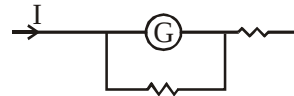
$$\Rightarrow n_1 : n_2 : n_3 : n_4 = \sqrt{1} : \sqrt{4} : \sqrt{9} : \sqrt{16} = 1 : 2 : 3 : 4$$

19.



Surface in yz plane will have flux only due to E_x
 $\phi = E_\perp A = (5)(20) = 100$

20.



$$G = \frac{G \cdot S}{G + S} + R \Rightarrow R = \frac{G^2}{G + S}$$

21. For isobaric process $\frac{V_2}{V_1} = \frac{T_2}{T_1} \Rightarrow V_2 = V_1 \times \left(\frac{T_2}{T_1} \right)$

$$\text{Increase} = \frac{(T_2 - T_1)V_1}{T_1} = V'$$

$$V' - V = \frac{V}{T} = \text{change in volume}$$

22. Rate of energy $\frac{Q}{t} = P = A\epsilon\sigma T^4 \Rightarrow P \propto T^4$

$$\Rightarrow \frac{P_1}{P_2} = \left(\frac{T_1}{T_2} \right)^4 = \left(\frac{927 + 273}{127 + 273} \right)^4 \Rightarrow P_1 = 405 \text{ W}$$

23. Mass per unit length $m = \frac{2 \times 10^{-4}}{0.5} \text{ kg/m} = 4 \times 10^{-4} \text{ kg/m}$

Frequency of 2nd harmonic $n_2 = 2n_1$

$$= 2 \times \frac{1}{2l} \sqrt{\frac{T}{m}} = \frac{1}{0.5} \sqrt{\frac{20}{4 \times 10^{-4}}} = 447.2 \text{ Hz}$$

24. $E_x \Rightarrow - \left(\frac{\partial v}{\partial x} \right) \hat{i}$

$$\Rightarrow - \left[- \frac{1000}{x^2} - \frac{3000}{x^3} - \frac{1500}{x^4} \right] \hat{i}$$

$$(E_x)_{x=1} \Rightarrow + [1000 + 3000 + 1500] \hat{i}$$

$$\Rightarrow 5500 \hat{i}$$

25. $C_1 = \frac{K_1 \epsilon_0 A}{d/2}$ $C_2 = \frac{K_2 \epsilon_0 A}{d/2}$

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} \Rightarrow C = \frac{2\epsilon_0 A K_1 K_2}{d(K_1 + K_2)}$$

26. Initially $\eta = \frac{T_1 - T_2}{T_1} \Rightarrow 0.5 = \frac{T_1 - (273 + 7)}{T_1}$

$\Rightarrow \frac{1}{2} = \frac{T_1 - 280}{T_1} \Rightarrow T_1 = 560 K$

Finally $\eta_1' = \frac{T_1' - T_2}{T_1'} \Rightarrow 0.7 = \frac{T_1' - (273 + 7)}{T_1'}$

$\Rightarrow T_1' = 933 K$

$\therefore$ increase in temperature
 $= 933 - 560 = 373 K \approx 380 K$

27. $n = \frac{\omega}{2\pi} = \frac{220}{2\pi} = 35 \text{ Hz}$

$v_{\max} = \omega a = 220 \times 0.30 \text{ m/s} = 66 \text{ m/s}$

28. Fundamental frequency $n = \frac{v}{2l}$

$\Rightarrow 350 = \frac{350}{2l} \Rightarrow l = \frac{1}{2} \text{ m} = 50 \text{ cm}$

29. $\frac{Q_1}{4\pi R^2} = \frac{Q_1 + Q_2}{4\pi \times 4R^2}$

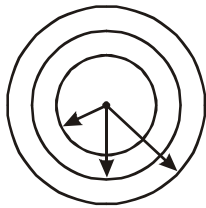
$= \frac{Q_1 + Q_2 + Q_3}{4\pi \times 9R^2}$

$4Q_1 \Rightarrow Q_1 + Q_2 \quad \dots(1)$

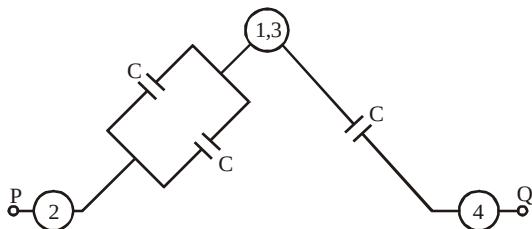
$9Q_1 \Rightarrow Q_1 + Q_2 + Q_3 \quad \dots(2)$

Solving

$Q_1 : Q_2 : Q_3 \Rightarrow 1 : 3 : 5$



30.



$C_{PQ} = \frac{2C \times C}{2C + C} = \frac{2C}{3} = \frac{2}{3} \frac{\epsilon_0 A}{d}$

31. Coefficient of performance

$K = \frac{T_2}{T_1 - T_2} = \frac{273}{303 - 273} = \frac{273}{30} = 9$

33. When the train is approaching the stationary observer frequency heard by the observer

$n' = \frac{v + v_0}{v} n$

when the train is moving away from the observer then frequency heard by the observer

$n'' = \frac{v - v_0}{v} n$

it is clear that n and n'' are constant and independent of time. Also $n' > n''$.

34. $q = i t$

$q = 5 \times 4 \times 60 = 1200 \text{ C}$

35. $C_{\text{series}} = \frac{C}{4} = C_1$

$C_{\text{parallel}} = 4C = C_2$

$\frac{C_1}{C_2} = \frac{1}{16}$

36. By adjoining graph $W_{AB} = 0$ and

$W_{BC} = 8 \times 10^4 [5 - 2] \times 10^{-3} = 240 \text{ J}$

$\therefore W_{AC} = W_{AB} + W_{BC} = 0 + 240 = 240 \text{ J}$

Now, $\Delta Q_{AC} = \Delta Q_{AB} + \Delta Q_{BC} = 600 + 200 = 800 \text{ J}$

From FLOT $\Delta Q_{AC} = \Delta U_{AC} + \Delta W_{AC}$

$\Rightarrow 800 = \Delta U_{AC} + 240 \Rightarrow \Delta U_{AC} = 560 \text{ J}$

37. $\frac{T'}{T} = \sqrt{\frac{g}{g' + a}} = \sqrt{\frac{g}{g + 5g}} = \sqrt{\frac{1}{6}} \Rightarrow T' = \frac{T}{\sqrt{6}}$

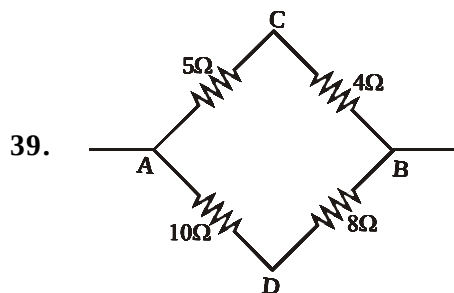
38. Initially potential difference across each capacitor

$V_1 = \frac{20}{(10 + 20)} \times 200 = \frac{400}{3} \text{ V}$

and $V_2 = \frac{10}{(10 + 20)} \times 200 = \frac{200}{3} \text{ V}$

Finally common potential $V = \frac{C_1 V_1 + C_2 V_2}{C_1 + C_2}$

$V = \frac{10 \times \frac{400}{3} + 20 \times \frac{200}{3}}{(10 + 20)} = \frac{800}{9} \text{ V}$



$$R_{AB} = \frac{9 \times 18}{27} = 6\Omega$$

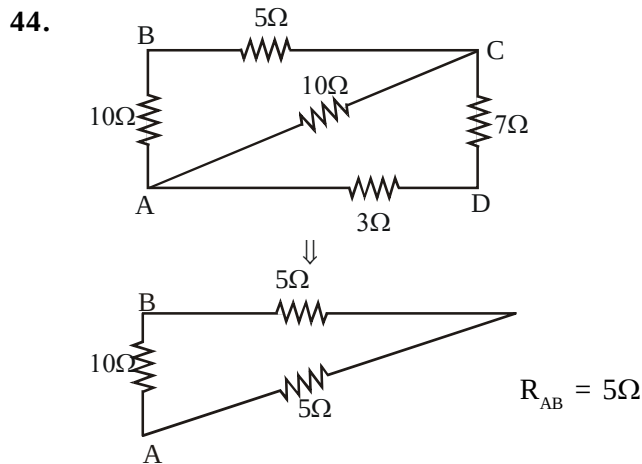
40. $I_{\max} = \frac{E}{R/2}$

$$50\% \text{ of } I_{\max} = \frac{E}{R}$$

when current through cell is $\frac{E}{R}$ then. No current will flow through capacitor at this time
 $q = CE$

- 42.** When external force is applied, one spring gets extended and another one gets contracted by the same distance hence force due to two springs act in same direction.

i.e. $F = F_1 + F_2 \Rightarrow -kx = -k_1x - k_2x \Rightarrow k = k_1 + k_2$



45. Total energy consumed = $\frac{60 \times 8}{1000} = 0.48 \text{ kWh}$

So cost = $0.48 \times 1.25 = 0.6$ Rs.

- 46.** Let VL of each are taken

$$n_{urea} = 0.1 \times V$$

$$n_{\text{Glucose}} = 0.1 \times V$$

$$\text{Total moles} = 0.1V + 0.1V$$

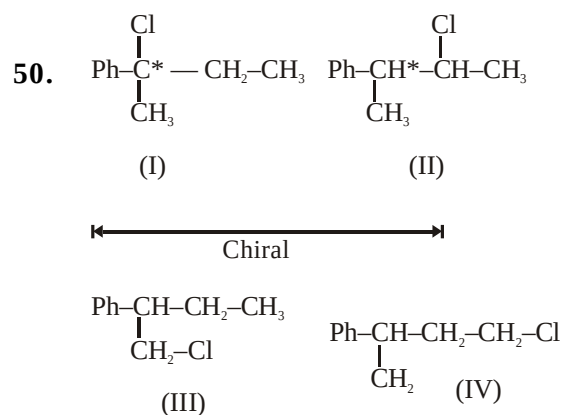
Total volume = $2V$

$$\text{Resultant Conc.} = \frac{0.2V}{V} = 0.1M$$

$$\begin{aligned} 47. \quad E_{\text{cell}} &= 0.0591 \log \frac{C_2}{C_1} \\ &= 0.0591 \log \frac{10^{-4}}{10^{-6}} \\ &= 0.0591 \log 10^2 \\ &= 0.0591 \times 2 \log 10 \\ &= 0.1182 \text{ Volt} \end{aligned}$$

- 49.** fact (NCERT)

reaction is aromatisation and takes place in presence of oxides of Cr, Mo or V in presence of Al_2O_3 .



- 51.** $i = 1 - \alpha + n\alpha$ [for dissociation]

$$i = 1 - \alpha + 3\alpha$$

$$\mathbf{i} = \mathbf{i} + 2\alpha$$

52. $r = k[A]^0$

$r = k \rightarrow$ It is independent of conc. but
k depends on temperature

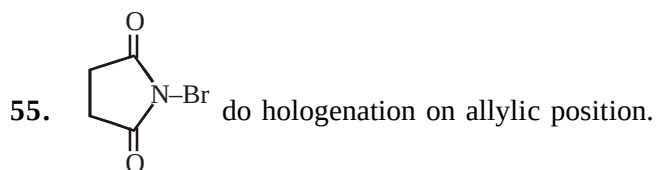
53. $k = A_0 e^{\frac{-40000}{T}}$

$$k = A \cdot e^{\frac{-E_a}{RT}}$$

$$\frac{E_a}{R} = 40000$$

$$\begin{aligned} E_a &= 40000 \text{ R} \\ &= 80000 \text{ Cal} \\ &= 80 \text{ kCal} \end{aligned}$$

54. Sulphonation is reversible and rate also depends on breaking of C-H bond. Heavier isotope forms stronger bond so takes max. time.



56. $\frac{\ln 2}{k_1} = 60$

$k_1 = \frac{\ln 2}{60}$

For zero order

$\frac{A_0}{2k_0} = 30$

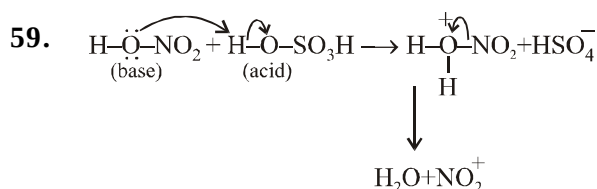
$k_0 = \frac{A_0}{60}$

$\therefore \frac{k_1}{k_0} = \frac{\ln 2}{A_0}$

$\frac{k_1}{k_0} = \frac{0.693}{1.386} = 0.5 \frac{\text{sec}^{-1}}{\text{mol} \cdot \text{dm}^{-3} \text{sec}^{-1}}$

$= 0.5 \text{ mol}^{-1} \text{dm}^3 \text{ :-}$

58. $\text{L.R.R} = \frac{r^+}{r^-} = \frac{126}{216} = 0.58$



60. In presence of strong base like  mechanism show $\text{S}_\text{N}2$ and product formed as Inversion in configuration.

61. $n_{\text{K}_2\text{SO}_4} = \frac{87}{174} \quad \text{K}_2\text{SO}_4 \quad i = 3$

$m = \frac{n}{w_{\text{solvent}}} \times 1000 = \frac{0.5}{500} \times 1000 = 1$

$\Delta T_f = i \times K_f \times m = 3 \times 1.8 \times 1 = 5.4$

$T_f = 0 - 5.4 = -5.4^\circ\text{C}$

$\Delta T_b = i \times K_b \times m = 3 \times 0.5 \times 1 = 1.5$

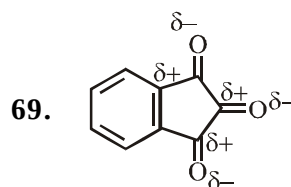
$T_b = 100 + 1.5 = 101.5$

62. $\uparrow T$; molecules becomes active. So they form chemical bond.

64. Reactivity in NAR depends on
1. electrophilic nature of carbonyl carbon
2. steric hindrance on carbonyl carbon
Ph is more EDG than CH_3

66. $m = \frac{\text{Molarity}}{\text{Density} - \frac{\text{Molarity} \times \text{Mol.mass}}{1000}}$

$m = \frac{9}{1.8 - \frac{9 \times 98}{1000}} = 9.8$



with attack on 2nd carbonyl group, the repulsion between same charges largely decreases.



$Q = \frac{[\text{Ni}^{+2}]^3}{[\text{Au}^{+3}]^2} = \frac{(0.01)^3}{(0.1)^2} = \frac{10^{-6}}{10^{-2}} = 10^{-4}$

$E_{\text{cell}} = E_{\text{cell}}^\circ - \frac{0.059}{6} \log 10^{-4}$

$E_{\text{cell}} = E_{\text{cell}}^\circ + \frac{4}{6} \times 0.059 \log 10$

$\therefore E_{\text{cell}} > E_{\text{cell}}^\circ$

72. $-\frac{1}{2} \frac{d[\text{NO}_2]}{dt} = \frac{[O_2]}{dt}$

$\Rightarrow \frac{-d[\text{NO}_2]}{dt} = 2 \times 1.5 \times 10^{-4} = 3 \times 10^{-4}$

$\Rightarrow \frac{-d[\text{NO}_2]}{dt} = k[\text{NO}_2]^2$

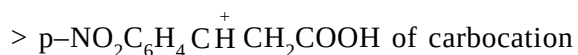
(from the unit of k, it is evident that it is second order reaction)

$\Rightarrow 3 \times 10^{-4} = 3 \times 10^{-2} [\text{NO}_2]^2$

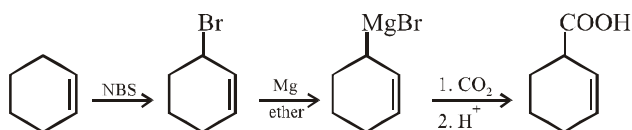
$\Rightarrow [\text{NO}_2]^2 = 10^{-2}$

$[\text{NO}_2] = 0.1 \text{ M}$

73. Reactivity in EAR $\propto$ stability of carbocation

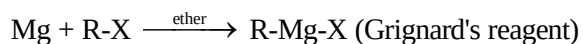


74.



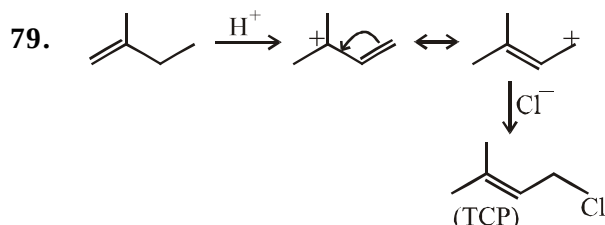
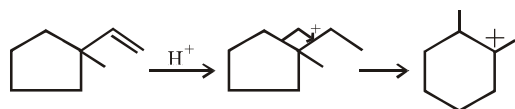
$$\begin{aligned} 76. \quad g_{\text{eq.}} \text{ of obtained Mg} &= \frac{Q}{96500} \\ &= \frac{9.65}{96500} = 10^{-4} \end{aligned}$$

$$\begin{aligned} \text{Moles of obtained Mg} &= \frac{g_{\text{eq}}}{n_{\text{factor}}} = \frac{10^{-4}}{2} = 0.5 \times 10^{-4} \\ &= 5 \times 10^{-5} \end{aligned}$$

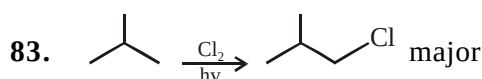


This, means; moles of R-Mg-X = moles of Mg = 5×10^{-5}

78. HCl does not show peroxide effect reaction is EAR and ring expansion takes place.



81. Conductivity of ion depends on hydration of ion.



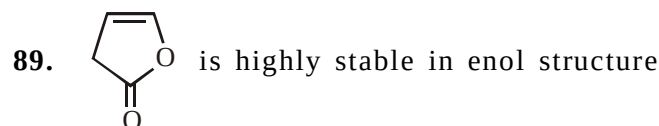
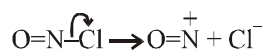
$$\text{Nu. of } 1^\circ\text{H (9)} \times \text{rate (1)} = 9$$

$$\text{Nu. of } 3^\circ\text{H (1)} \times \text{rate (5)} = 5$$

86.

$$\begin{aligned} \Lambda_{\text{eq}} &= \frac{1}{R} \times \left(\frac{\ell}{A} \right) \times \frac{1000}{N} \\ &= \frac{1}{200} \times 2 \times \frac{1000}{0.1} = 100 \text{ S cm}^2\text{eq}^{-1} \end{aligned}$$

88. This is EAR and $\text{NO}^{\oplus}$ acts as electrophile.



because of aromatic character.

91. NCERT-XI, Pg. # 101, Para-3(E), Para-4(H)

92. NCERT-XI, Pg. # 269

93. NCERT-XI, Pg. # 299

94. NCERT, Pg. # 51

96. NCERT-XI, Pg. # 108

97. NCERT, Pg # 129

98. NCERT, Pg # 129

99. NCERT, Pg # 59

Increasing number of people in reproductive age group celro causes population explosion.

100. NCERT-XII, Pg # 22(E), 23(H)

101. NCERT-XI, Pg. # 103, Para-2

102. NCERT-XI, Pg. # 270

103. NCERT-XI, Pg. # 296

104. NCERT, Pg. # 52

106. NCERT-XI, Pg. # 107

107. NCERT, Pg # 131

108. NCERT, Pg # 43,44

Bartholin gland found in female

109. NCERT, Pg # 59,60

Multiload-375 is copper releasing IUD

110. NCERT-XII, Pg # 28(E), 29(H)

111. NCERT-XI, Pg. # 103, Para-3

112. NCERT-XI, Pg. # 271

113. NCERT-XI, Pg. # 294

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| <p>115. NCERT, Pg. # 330</p> <p>116. NCERT-XI, Pg. # 112</p> <p>117. NCERT, Pg # 130</p> <p>118. NCERT, Pg # 45
Clitoris is made up of two corpora cavernosa</p> <p>119. NCERT, Pg # 62
45-50 million MTP all over world, MTP act-1971</p> <p>120. Module-1, Pg # 114(E), 113(H)</p> <p>121. NCERT-XI, Pg. # 103, Para-2, Fig. 7.4(b)</p> <p>122. NCERT-XI, Pg. # 275</p> <p>123. NCERT, Pg. # 47</p> <p>124. NCERT, Pg. # 55</p> <p>125. NCERT, Pg. # 333</p> <p>126. NCERT-XI, Pg. # 309</p> <p>127. NCERT, Pg # 141, 142</p> <p>128. NCERT, Pg # 48,49
Oogenesis already started during foetal life</p> <p>129. NCERT, Pg # 62</p> <p>130. Module-1, Pg # 112(E)(H)</p> <p>131. NCERT-XI, Pg. # 106, Para-2(E), Para-3(H)</p> <p>132. NCERT-XI, Pg. # 280</p> <p>133. NCERT, Pg. # 46</p> <p>136. NCERT-XI, Pg. # 312</p> <p>137. NCERT, Pg # 127</p> <p>138. NCERT, Pg # 49,50
In general oxytocin do not have role in regulation of menstrual cycle.</p> <p>139. NCERT, Pg # 60,64
Norplant effective upto 5 years Cu-7-copper releasing IUD, ICSI-In-vitro fertilisations</p> <p>140. NCERT-XII, Pg # 36(E), 38(H)</p> <p>141. NCERT-XI, Pg. # 258</p> <p>142. NCERT-XI, Pg. # 282</p> <p>143. NCERT, Pg. # 47</p> | <p>144. NCERT, Pg. # 53</p> <p>145. NCERT, Pg. # 336</p> <p>146. NCERT-XI, Pg. # 312</p> <p>147. NCERT, Pg # 127</p> <p>148. NCERT, Pg # 49,50
Ovulation 14th day
Rise in progesterone level → after ovulation means from 15 day - 25 day
Endometrium degeneration → 1-5th day</p> <p>149. NCERT-XII, Pg # 24</p> <p>150. Module-1, Pg # 119(E), 119(H)</p> <p>151. NCERT-XI, Pg. # 259</p> <p>152. NCERT-XI, Pg. # 285</p> <p>156. NCERT, Pg # 133</p> <p>157. NCERT, Pg # 137</p> <p>158. NCERT, Pg # 51,52</p> <p>159. NCERT-XII, Pg # 26</p> <p>160. Module-1, Pg # 119(E), 119(H)</p> <p>161. NCERT-XI, Pg. # 262</p> <p>162. NCERT-XI, Pg. # 288</p> <p>166. NCERT, Pg # 133</p> <p>167. NCERT, Pg # 137</p> <p>168. NCERT, Pg # 53,54</p> <p>169. Module-1, Pg # 99(E), 98(H)</p> <p>170. NCERT-XII, Pg # 26(E), 26(H), Figure-2.7(d)</p> <p>171. NCERT-XI, Pg. # 262</p> <p>172. NCERT-XI, Pg. # 298</p> <p>173. NCERT, Pg. # 50</p> <p>174. NCERT, Pg. # 316</p> <p>175. NCERT, Pg. # 325</p> <p>176. NCERT, Pg # 128</p> <p>177. NCERT, Pg # 138</p> <p>178. NCERT, Pg # 57,58</p> <p>179. Module-1, Pg # 106-107-108(E), 107-108(H)</p> <p>180. Module-1, Pg # 115-116(E), 116(H)</p> |
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